LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Marking of different wires in aircraft
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Marking of different wires in aircraft.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of wire markings on sample harness; Safety: Avoid pulling live wires; Supervision: Instructor verifies color code accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of wire marking in aircraft wiring?
(b) List two common causes of wiring degradation.
(c) Why is color coding essential for maintenance personnel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Standard color coding for wire identification • Marking methods for different wire sizes • Importance of legible and durable markings • Consequences of incorrect wire identification • Relationship between marking and circuit tracing	L & Ds	PPT, IB	15 min
(b)	• Causes of wiring degradation due to environmental factors • Impact of improper marking on maintenance efficiency • Techniques for preserving wire markings during repair • Common errors in wire marking and their risks • Role of markings in troubleshooting electrical faults	L & Ds	PPT, PDF	20 min
(c)	• Practical identification of marked wires on sample harness • Verification of color codes against standard charts • Documentation of observed marking conditions • Reporting discrepancies in wire identification • Review of best practices for wire marking	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how incorrect wire marking contributes to wiring degradation.
(b) Describe the steps to verify wire markings during a maintenance check.

12. Home Assignment and Follow Up	Prepare a brief note on how proper wire marking facilitates the transition to LP 17 on grounding methods.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of grounding, various types of grounding, grounding methods.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min